

# Buchanan Kerswell

*Curriculum Vitae: Oct 01 2026*

✉ [b.kerswell@liverpool.ac.uk](mailto:b.kerswell@liverpool.ac.uk)



I study how Earth's mantle and tectonic plates move using fieldwork, petrography, numerical geodynamic modeling, applied statistics, and various analytical techniques. My work focuses on rock formation and deformation at small scales, and how these processes affect larger systems. I work in remote mountains, use electron beams in labs, and run simulations on large computer clusters with my colleagues and students. I also teach, create learning materials, and explore how AI can improve geoscience.

## EDUCATION

May 2022 : Aug 2015 **PhD Geosciences** Boise State University

Supervisor: Dr. Matthew Kohn

Dissertation: Computational Approaches to Understanding Subduction Zone Geodynamics, Surface Heat Flow, and the Metamorphic Rock Record

Jul 2015 : Aug 2011 **BSc Earth Science** Utah Valley University (NSF Fellow)

Supervisor: Dr. Steven Fellows

Thesis: Geochemical Analysis Of Tourmaline as a Tool for Determining the Location and Characteristics of the Main Central Thrust, Central Nepal

## PROFESSIONAL APPOINTMENTS

present : Nov 2024 **Postdoc** University of Liverpool

Investigated phase transformations in mantle flow models under non-hydrostatic stress conditions

Oct 2024 : May 2023 **Postdoc** Géosciences Montpellier

Applied machine learning (ML) and artificial intelligence (AI) techniques to geochemical and thermodynamic datasets to increase the efficiency of predicting rock properties in numerical geodynamic simulations

May 2023 : Aug 2022 **Visiting Assistant Professor** Miami University

Adapted, updated, delivered, and graded course materials, mentored independent undergraduate research on high-pressure metamorphic rocks within the AUGITE program, taught graduate/undergraduate seminars on scientific communication including technical writing workshops and student-led public presentations and discussions, completed a New Faculty Teaching Enhancement Program with Miami's Center for Teaching and Learning, participated in cross-disciplinary events across campus

- May 2019 : Jan 2019 **Instructor** Boise State University  
Adapted, updated, delivered, and graded course materials, transitioned a field-based course to online-learning, produced a virtual field trip during the early COVID-19 pandemic
- Aug 2018 : Jun 2018 **Visiting Student** Sorbonne Université  
Supervisor: Dr. Philippe Agard  
Studied rheologic theory and application, implemented metamorphic-rheologic processes into numerical geodynamic models
- Dec 2017 : Sep 2017 **Visiting Student** ETH-Zürich  
Supervisor: Dr. Taras Gerya  
Studied computational fluid dynamic theory and application, practiced high performance computing, and developed finite-difference numerical geodynamic models of subduction
- May 2022 : Aug 2015 **Graduate Research Associate** Boise State University  
Supervisor: Dr. Matthew Kohn  
Studied petrologic theory and application, practiced chemistry- and physics-based analytical techniques, sampled various rock types and tectonic structures from exhumed HP terranes (Catalina Island, W Alps, NW India, Greece), led seminars on tectonic theory and application
- May 2021 : Aug 2015 **Graduate Research Associate** Boise State University  
Adapted, updated, delivered, and graded course materials, organized, ran, and graded up to one field exercise per week, developed web pages and inclusive formats to enhance online-learning experiences
- May 2021 : Aug 2015 **Guest Lecturer** Boise State University  
Introduced finite-difference numerical modeling approaches, led students through live coding tutorials (a student's code featured in [this peer-reviewed publication](#)), presented selected material on subduction zone metamorphism and geodynamics

## PEER-REVIEWED PUBLICATIONS

### BOOK CHAPTERS

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- [1] **B. Kerswell** (2026). Methods in Tectonics: From Big Data to Machine Learning. *Geoscience in Practice: Geological Society of London*. [In Review]

### JOURNAL ARTICLES

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- [9] **B. Kerswell**, M. Kohn (2026). Along-Strike Heat Flow Continuity Near Subduction Zones Is Mostly Ambiguous, Not Globally Uniform. *JGR: SE*. [In Review] [repo](#)

- [8] **B. Kerswell**, J. Wheeler, R. Gassmöller, J. Huw Davies, I. Papanagnou, S. Cottaar, D. P. Dobson (2026). Beyond Equilibrium II: Ringwoodite Decomposition Kinetics Amplify 660 km Discontinuity Depressions and Offer an Independent Control on Slab Stagnation. *JGR: SE*. [In Review] [repo](#)
- [7] **B. Kerswell**, J. Wheeler, R. Gassmöller, J. Huw Davies, I. Papanagnou, S. Cottaar (2026). Beyond Equilibrium: Kinetic Thresholds and Rheological Feedbacks Create a Potentially Complex 410 in Slab Regions. *JGR: SE*. doi: 10.1029/2026JB033781 [online](#) [pdf](#) [repo](#) [google](#)
- [6] **B. Kerswell**, N. Cerpa, A. Tommasi, M. Godard, J. Alberto Padrón-Navarta (2024). RocMLMs: Predicting Rock Properties through Machine Learning Models. *JGR: MLC*. doi: 10.1029/2024JH000264 [online](#) [pdf](#) [repo](#) [google](#)
- [5] **B. Kerswell**, M. Kohn, T. Gerya (2023). Computing Rates and Distributions of Rock Recovery in Subduction Zones. *Geochemistry, Geophysics, Geosystems*. doi: 10.1029/2022GC010834 [online](#) [pdf](#) [repo](#) [google](#)
- [4] M. Klöcking, L. Wyborn, K. Lehnert, B. Ware, A. Prent, L. Profeta, F. Kohlmann, W. Noble, I. Bruno, S. Lambart, H. Ananuer, N. Barber, H. Becker, M. Brodbeck, H. Deng, K. Deng, K. Elger, G. de Souza Franco, Y. Gao, K. Ghasera, D. Hezel, J. Huang, **B. Kerswell**, J. Kim, H. Koch, A. Lanati, G. ter Maat, N. Martínez-Villegas, L. Nana Yobo, N. Randazzo, A. Redaa, W. Schäfer, M. R. Swing, R. Taylor, M. Traun, J. Whelan, T. Zhou (2023). Community Recommendations for Geochemical Data, Services and Analytical Capabilities in the 21st Century. *Geochimica et Cosmochimica Acta*. doi: 10.1016/j.gca.2023.04.024 [online](#) [pdf](#) [google](#)
- [3] **B. Kerswell**, M. Kohn, T. Gerya (2021). Backarc Lithospheric Thickness and Serpentine Stability Control Slab-Mantle Coupling Depths in Subduction Zones. *Geochemistry, Geophysics, Geosystems*. doi: 10.1029/2020GC009304 [online](#) [pdf](#) [repo](#) [google](#)
- [2] S. Long, M. Kohn, **B. Kerswell**, J. Starnes, K. Larson, N. Blackford, E. Soignard (2020). Thermometry and Microstructural Analysis Imply Protracted Extensional Exhumation of the Tso Moriri UHP Nappe, Northwestern Himalaya: Implications for Models of UHP Exhumation. *Tectonics*. doi: 10.1029/2020TC006482 [online](#) [pdf](#) [repo](#) [google](#)
- [1] M. Kohn, A. Castro, **B. Kerswell**, C. Ranero, F. Spear (2018). Shear Heating Reconciles Thermal Models with the Metamorphic Rock Record of Subduction. *Proceedings of the National Academy of Sciences*. doi: 10.1073/pnas.1809962115 [online](#) [pdf](#) [google](#)

## CONFERENCE PROCEEDINGS

- [12] **B. Kerswell**, J. Wheeler, R. Gassmöller, J. Huw Davies, I. Papanagnou, S. Cottaar (2026). **\*Invited** Beyond Equilibrium: Kinetic Thresholds and Rheological Feedbacks Create a Potentially Complex 410 in Slab Regions. *European Geosciences Union 2026, Vienna Austria*. doi: 10.5194/egusphere-egu26-6886 [abstract](#) [slides](#)
- [11] **B. Kerswell**, N. Cerpa, A. Tommasi, M. Godard, J. Alberto Padrón-Navarta (2024). **\*Invited** RocMLMs: Predicting Rock Properties through Machine Learning Models. *American Geophysical Union 2024, Washington D.C.*. [abstract](#) [slides](#) [video](#)

- [10] **B. Kerswell**, N. Cerpa, A. Tommasi, M. Godard, J. Alberto Padrón-Navarta (2024). RocMLMs: Predicting Rock Properties through Machine Learning Models. *European Geosciences Union 2024, Vienna Austria*. doi: 10.5194/egusphere-egu24-8578 [abstract](#) [slides](#)
- [9] **B. Kerswell**, M. Kohn (2023). A Comparison of Surface Heat Flow Interpolations Near Subduction Zones. *Goldschmidt 2023, Lyon France*. [slides](#)
- [8] D. Sims, **B. Kerswell** (2023). Deformation of Dry High-Pressure Eclogites During Tectonic Slicing of Subducted Oceanic Lithosphere: a Case Study from the Monviso Ophiolite, Italy. *Geological Society of America North-Central 2023, Grand Rapids*. doi: 10.1130/abs/2023NC-386862 [abstract](#)
- [7] C. Morrison, **B. Kerswell** (2023). Comparing PT Paths of Metamorphic Rocks Determined by Quantitative and Semi-Quantitative Approaches: a Case Study from the Monviso Ophiolite, Italy. *Geological Society of America North-Central 2023, Grand Rapids*. doi: 10.1130/abs/2023NC-386686 [abstract](#)
- [6] **B. Kerswell**, M. Kohn (2022). A Comparison of Surface Heat Flow Interpolations Near Subduction Zones. *American Geophysical Union 2022, Chicago*. [abstract](#) [slides](#) [video](#)
- [5] **B. Kerswell**, M. Kohn, T. Gerya (2022). Computing Rates and Distributions of Rock Recovery in Subduction Zones. *Geological Society of America 2022, Denver*. [abstract](#) [slides](#)
- [4] **B. Kerswell**, M. Kohn, T. Gerya (2022). Computing Rates and Distributions of Rock Recovery in Subduction Zones. *Goldschmidt 2022, Honolulu USA*. [abstract](#) [slides](#) [video](#)
- [3] **B. Kerswell**, M. Kohn, T. Gerya (2018). A Physical Mechanism Explaining the Common Depth of Slab-Mantle Coupling and Formation of a Rheologic Backstop at ~80 km Depth. *Goldschmidt 2018, Boston USA*. [abstract](#)
- [2] **B. Kerswell**, M. Kohn, T. Gerya (2018). Backarc Lithospheric Thickness and Serpentine Stability Control Slab-Mantle Coupling Depths in Subduction Zones. *50 Years of Plate Tectonics: Then, Now, and Beyond, Collège de France, Paris France*.
- [1] **B. Kerswell**, S. Fellows, S. Emerman, S. Panday, S. Adhikari (2014). Geochemical Analysis Of Tourmaline as a Tool for Determining the Location and Characteristics of the Main Central Thrust, Central Nepal. *Geological Society of America 2014, Vancouver Canada*. [abstract](#) [slides](#)

## INVITED TALKS

- [6] **B. Kerswell**, J. Wheeler, R. Gassmöller, J. Huw Davies, I. Papanagnou, S. Cottaar (2026). Beyond Equilibrium: Kinetic Thresholds and Rheological Feedbacks Create a Potentially Complex 410 in Slab Regions. *European Geosciences Union 2026, Vienna Austria*. [slides](#)
- [5] **B. Kerswell** (2026). Beyond Equilibrium: Reaction Kinetics in Geodynamic Models. *ASPECT User Meeting 2026*. [slides](#)
- [4] **B. Kerswell**, J. Wheeler, R. Gassmöller (2025). Displaced and Faded: how thermodynamics and kinetics collude to complicate seismic structures in Earth's mantle. *University of Liverpool, Earth Science Research Group*. [slides](#) [video](#)

- [3] **B. Kerswell**, N. Cerpa, A. Tommasi, M. Godard, J. Alberto Padrón-Navarta (2024). RocMLMs: Predicting Rock Properties through Machine Learning Models. *American Geophysical Union 2024, Washington D.C.* [slides](#) [video](#)
- [2] **B. Kerswell** (2022). Evaluating Distributions of High-Pressure Rock Recovery in Subduction Zones with Large Empirical Datasets and Numerical Simulations. *University of Minnesota.* [slides](#)
- [1] **B. Kerswell** (2022). Evaluating Aspects of Geodynamic Uniformity Among Subduction Zones with Large Empirical Datasets and Numerical Simulations. *University of Southern California.* [slides](#) [video](#)

## SOFTWARE

### LEAD DEVELOPER

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- [4] **RxnDB**  Python (MIT License)  
A mineral reaction database toolkit
- [3] **RocMLM**  Python (MIT License) doi: 10.17605/OSF.IO/K23TB  
A toolkit for predicting rock properties through machine learning models
- [2] **Peakfit Explorer**  R (MIT License) doi: 10.17605/OSF.IO/2BU3C  
An app for performing Gaussian Mixture Modeling on 1D data
- [1] **Light Slides**  Website (MIT License)  
A set of thin sections sampled from Matt Kohn's personal library

### CONTRIBUTOR

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- [2] **Candi**  bash (LGPL-3.0 License)  
Downloads, configures, builds and installs deal.II
- [1] **ASPECT**  c++ (GPLv2 License)  
A finite element code to simulate convection in both 2D and 3D models

## PROFESSIONAL DEVELOPMENT

### WORKSHOPS

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- Jul 2026      **ASPECT Hackathon** Liensfeld, Germany  
Implemented a new reaction kinetics framework in ASPECT that enables sequences of phase transformations governed by kinetic models

- Jun 2025 **ASPECT Hackathon** Garden City, Utah, USA  
Implemented a new material model in ASPECT that governs phase transformation kinetics via non-equilibrium thermodynamics
- Apr 2025 **Research Integrity Training** University of Liverpool  
Evaluated my own research practice in terms of research integrity and ethics. Lead by Dr. Roberto Ferrero.
- Jul 2020 **Preparing for an Academic Career** Earth Educators Rendezvous  
Discussed career paths, prepared and reviewed job application materials
- Feb 2016 **SIMS Student Workshop** UCLA EPSS  
Engaged in a 5-day workshop focused on principles and practice of secondary-ion mass spectrometry (SIMS)

## SYMPOSIA

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- Jun 2026 **Ada Lovelace** Šéc, Czechia  
Presented research and engaged in networking and scientific discussions focusing on numerical geodynamic modeling
- Apr 2025 **Metamorphic Studies Group** Liverpool, United Kingdom  
Presented research and engaged in networking and scientific discussions focusing on metamorphic petrology
- Apr 2025 **Imaging Planetary Solid Dynamics** Cambridge, United Kingdom  
Presented research and engaged in networking and scientific discussions focusing on seismology and geodynamics
- Sep 2024 **Ada Lovelace** Sète, France  
Presented research and engaged in networking and scientific discussions focusing on numerical geodynamic modeling
- Jul 2024 **ERC RhEoVOLUTION** Montpellier, France  
Presented research and engaged in networking and scientific discussions focusing on deformation in the solid Earth
- Jul 2024 **French Network on Subduction Zones (FrenSZ)** Barcelonnette, France  
Presented research and engaged in networking and scientific discussions focusing on subduction zone dynamics
- Jan 2024 **ERC RhEoVOLUTION** Villard-de-Lans, France  
Presented research and engaged in networking and scientific discussions focusing on deformation in the solid Earth
- May 2023 **Réseau Numérique en Terre Solide (NuTS)** Université Lyon 1  
Presented research and engaged in networking and scientific discussions focusing on current capabilities and new directions for scientific computing in the geosciences

- Jul 2022 **Earth Science Meets Data Science** Goldschmidt Conference Honolulu  
Explored scientific challenges in geochemistry, showcased examples of existing data solutions/infrastructures/services, discussed recommendations, best practices, and essential features of a globally standardised geochemical data framework
- Feb 2019 **GeoPRISMS Synthesis & Integration TEI** Hotel Menger, San Antonio TX  
Engaged in community efforts to synthesize the successful achievements of the MARGINS and GeoPRISMS programs, participated in an Early-Career Investigator symposium to identify outstanding questions in subduction zone research
- Aug 2018 **GeoPRISMS EFIRE** Boston College, MA  
Engaged in community building, initiated strategic sharing and storage protocols for EFIRE samples, presented research on subduction zone mechanics, planned a field institute for 2019
- Dec 2016 **GeoPRISMS EFIRE** Marin Headlands Hostel, CA  
Engaged in community building, discussed past research to identify future targets for subduction zone investigations, planned a field institute for 2017

## TEACHING & LEARNING

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- Aug 2022 **New Faculty Teaching Enhancement Program** Miami University CTE  
Engaged in numerous activities on improving scholarly teaching
- Aug 2020 **Exploration in Pedagogy** Boise State University CTL  
Engaged in numerous activities on pedagogy and inclusive teaching
- Aug 2020 **Identity Formation Through Teaching** Boise State University CTL  
Planned K-12 lessons with Education majors based on next generation science standards

## AWARDS & HONORS

### COLLABORATIVE GRANTS

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- Mar 2022 **Postdoc** £2.8M NERC NE/V018477/1  
Feedbacks between mineral reactions and mantle convection
- Nov 2020 **Postdoc** €2.5M ERC 882450  
RhEoVOLUTION: Micro-scale dependent, time- and space-evolving rheologies
- Jun 2016 **PhD Student** \$4.0M NSF 1545903  
ExTerra Field Institute

### FELLOWSHIPS

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- Sep 2026 **PI** £2.18M Royal Society University Research Fellowship [Pending]  
AI-Driven Critical Mineral Exploration: From Grain-Scale Analysis to Crustal-Scale Deposit Formation
- Jun 2026 **PI** £1.58M NERC Independent Research Fellowship [Pending]  
AI-Driven Critical Mineral Exploration: From Grain-Scale Analysis to Crustal-Scale Deposit Formation
- Apr 2026 **PI** £350k UKRI AI Metascience Early Career Fellowship [Not funded]  
AI-Assisted Microscopy in Geoscience: A Practical Metascientific Investigation of Changing Research Practices, Skills, and Human Roles
- Nov 2024 **PI** €89k Humboldt Foundation Fellowship [Not funded]  
Interplay between Metamorphic Reactions and Fluid-Mediated Strain Localization: A Comparative Study of Mélange-Type and Coherent-Type Ophiolites in the Cycladic Archipelago and Western Alps
- Feb 2024 **PI** €196k Marie Curie Fellowship (score: 85.6/100) [Not funded]  
SMARTGEOS: Supervised Machine learning for Accurate and Rapid Thermodynamics in GEodynamic Simulation
- Aug 2011 **PI** \$40k NSF Undergraduate STEM Fellowship  
Engaged in interdisciplinary activities with faculty mentors and undergraduates from STEM fields

## MINOR AWARDS

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- Jun 2019 **Research Grant** \$3,500 GeoPRISMS: EFIRE  
Sampling mafic eclogites from Monviso, Italy
- Feb 2016 **Research Grant** \$1,500 Geological Society of America  
Zr-in-Rutile EPMA analysis

## HONORS

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- May 2021 **Outstanding Graduate Teaching Assistant** Boise State University  
Voted best Teaching Assistant by the Department of Geosciences student body
- May 2015 **Outstanding Undergraduate Contributions** Utah Valley University  
Engaged in community building, organized field trips, and served as president of the Geology Club

## PROFESSIONAL SERVICE

I regularly serve as an expert reviewer for top-tier international journals, including *Earth and Planetary Science Letters*, *Journal of Geophysical Research*, *Geophysical Journal International*, *Geochemistry*, *Geophysics*, *Geosystems*, *Journal of Metamorphic Geology*, *Scientific Reports*, and *iScience*. I frequently review research proposals via major international funding agencies, in-

cluding the National Science Foundation (NSF) and the German Research Foundation (DFG). I also contribute to academic governance by co-organizing and chairing conference sessions and symposia for international organizations, as well as volunteer judging of student presentations for major societies such as the American Geophysical Union (AGU).

- Jun 2026      **Faculty Host** Year 12 Outreach  
60+ participants  
Co-organized and hosted student recruiting event involving analog simulations of volcanic eruptions
  
- Apr 2025      **Organizing Committee** MinSoc Metamorphic Studies Group  
50+ participants  
Co-organized and hosted meeting and workshop on behalf of the Mineralogical Society
  
- May 2023      **Faculty Judge** Road Test Pitch Competition 2023  
100+ undergraduates  
Heard undergraduates pitch early-stage business ideas that they developed at Miami University's Farmers Business School
  
- Mar 2023      **Faculty Mentor** Social Innovation Weekend 2023  
100+ undergraduates  
Engaged with student teams at Miami University's Farmers Business School as they proposed innovative new technologies regarding climate change and its impacts on our society

## PROFESSIONAL AFFILIATIONS

I maintain an active, long-standing membership in leading international professional societies within the geosciences, including the American Geophysical Union (AGU), European Geosciences Union (EGU), Geological Society of America (GSA), Mineralogical Society of America (MSA), and International Association for Mathematical Geosciences (IAMG).

## TEACHING

The table below shows my comparative scores versus the departmental average for the following prompt given to students at the end of term: "Upon reflection, this instructor is an effective teacher" (out of 4 = "strongly agree").

| term | course  | title                               | format      | role       | size | score | dpt   |
|------|---------|-------------------------------------|-------------|------------|------|-------|-------|
| S23  | GLG 556 | Environmental & Economic Mineralogy | Seminar     | Instructor | 7    | 3.9/4 | 3.4/4 |
| S23  | GLG 456 | Environmental & Economic Mineralogy | Seminar     | Instructor | 30   | 3.9/4 | 3.4/4 |
| S23  | GLG 111 | Dynamic Earth                       | Lecture/Lab | Instructor | 342  | 3.6/4 | 3.4/4 |
| F22  | GLG 211 | Chemistry of Earth Systems          | Lecture/Lab | Instructor | 27   | 3.4/4 | 3.4/4 |

| term | course   | title                                    | format      | role                | size | score | dpt   |
|------|----------|------------------------------------------|-------------|---------------------|------|-------|-------|
| F22  | GLG 121  | Environmental Geology                    | Lecture/Lab | Instructor          | 196  | 3.7/4 | 3.5/4 |
| S21  | GEOS 345 | Igneous & Metamorphic Petrology          | Lecture/Lab | TA                  | 23   | —     | —     |
| F20  | GEOS 471 | Field Methods                            | Field Lab   | TA                  | 8    | —     | —     |
| F20  | GEOS 315 | Sedimentology & Stratigraphy             | Lecture/Lab | TA                  | 21   | —     | —     |
| S20  | GOES 314 | Structural Geology                       | Lecture/Lab | Instructor          | 20   | —     | —     |
| S19  | GEOS 645 | Physics & Chemistry of Mountain Building | Seminar     | Guest lecturer      | 12   | —     | —     |
| S18  | GEOS 597 | Subduction Zone Processes                | Seminar     | Co-instructor       | 15   | —     | —     |
| S16  | GEOS 300 | Earth Materials                          | Lecture/Lab | TA & guest lecturer | 23   | —     | —     |
| F15  | GEOS 101 | Introductory Geology                     | Lecture/Lab | TA                  | 48   | —     | —     |

## MENTORING

| year        | name        | role          | thesis                                        |
|-------------|-------------|---------------|-----------------------------------------------|
| 2023 : 2022 | C. Morrison | Undergraduate | <u>Comparing PT paths: Monviso Ophiolite</u>  |
| 2023 : 2022 | D. Sims     | Undergraduate | <u>Strain localization: Monviso Ophiolite</u> |